A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 24

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1** Harvest rate below the maximum sustainable yield – only economically valuable species (mahogany) selectively logged, not clear-felled **1** Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing – large seed trees and small trees left to grow **1** Buffer zones with no harvesting or machinery near rivers / wetlands **1** Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees **1** Each felled tree tagged and tracked to the retailer for FSC accreditation – rangers control illegal logging – tree nurseries replant over-exploited and ecologically important species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it **1** Employment as rangers, forestry and tourism workers – ecotourists use local restaurants, produce and souvenirs – offcuts given to local carvers **1** Benefits give the community an incentive to protect the forest – carbon sequestration payments from companies fund management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept – timber resource used over a longer period **1 Disadvantage:** more labour-intensive and expensive – more tracks needed per tree **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps **1** Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
02.2	**1 mark each** 1 mark for each of the 4 points below. **1** Fixed mooring buoys so boats do not anchor on coral **1** Education of divers / snorkellers not to touch coral or kick up sediment – bans on spear fishing and souvenir collection **1 Fishing controls:** no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling **1** Protection of turtle nesting beaches – control of coastal development – protection of mangroves and seagrass – artificial reefs (reef balls, sunk ships / oil rigs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** No sunlight so no symbiotic algae – polyps rely only on captured food so have far less energy and grow very slowly **1** Damage from trawls takes hundreds of years to repair – the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Deep-water trawling (also by-catch) – oil and gas exploration in deeper water – ocean acidification **1** Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands – HEP schemes alter river levels flooding sandbank nests of turtles / birds – species with narrow tolerance (specialists) affected first – sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade **2** Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae **2** Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Reduces light penetration so submerged plants cannot photosynthesise **1** Blocks the gills of filter feeders such as bivalve molluscs – sediment smothers the bed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Rare species often have very specialised habitat requirements / narrow ranges of tolerance **1** A small change moves conditions outside their range while generalists survive – small populations cannot recover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Readings drift so they may not be accurate – calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions **1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly **1** If a meter cannot be adjusted, record the percentage error and correct the results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Same height above the ground **1** No obstacles to air flow – impeller type facing into the wind **1** Axis of rotation horizontal (impeller) or vertical (cup) – same time period / averaging **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling **1** Evaporation cools the wet bulb more when humidity is low – the temperature difference is compared with a table to find humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Test strips with reagent bands compared with a colour chart (semi-quantitative) **1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength **1** Ion-selective electrode measuring electrode potential (other ions can interfere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1
05.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion – waste products of one process are raw materials of another so wastes do not build up **1** Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere **1** Allowing more plant growth, further stabilising climate and soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Lack of ancient historical / instrumental data **1** Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions **1** Lack of sophisticated equipment / inability to measure many factors **1** Limited coordination between researchers – lack of data collection in many areas (oceans, poles) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
06.3	**1 mark each** 1 mark for each of the 4 points below. **1** The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell **1** Air bubbles trapped in ice give the composition of the past atmosphere (CO2, methane) **1** Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) **1** More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** All greenhouse gases expressed as CO2 equivalent (allowing for different warming strengths) **1** Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) **1** A fair share of the infrastructure: roads, car parks, signals, street lighting **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels **1** Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
08.3	**1 mark each** 1 mark for each of the 5 points below. **1** Hardy, disease-resistant Boran cattle – regular tick dipping reduces ticks for wild herbivores too **1** Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth **1** Electric fences keep game in and poachers out – biological corridors let animals (except rhinos) move to other reserves – problem elephants relocated **1 Monitoring:** radio-collared lions, drones, camera traps, track surveys – endangered species (Grevy's zebra, northern white rhino) protected **1 Community links:** ecotourism income funds health, education and water projects – local employment – cattle breeding help for local farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Other MEDCs (USA, Canada, Australia) also had surpluses – Japan already had suppliers **1** Eastern Europe / USSR needed food but could not pay **1** Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 4 points below. **1** Quotas limiting production (eg milk) **1** Set-aside payments for taking land out of production **1** Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) **1** Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
09.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape **1** Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat **1** Beetle banks of tussocky grass - natural pest predators **1** Skylark nesting plots – wild bird seed / nectar plants - winter food and pollinators **1** In-field tree protection – controlled stocking density to reduce compaction – ditch management – dry-stone walls – woodland management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** By-catch = 5.9 + 1.9 = 7.8 t – total = 8.4 t **1** 7.8 / 8.4 x 100 = 92.9% (accept 93%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
10.2	1 mark for the point below. **1** The nets have a very small mesh size so few non-target animals can escape **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mesh size** → net hole size, set so that young fish escape	1
10.3	**1 mark each** 1 mark for each of the 3 points below. **1** They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young **1** They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred **1** Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young – they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Impacts:** population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • **Catch quotas:** limit weight landed – work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • **Gear design:** mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag) – turtle exclusion devices – acoustic pingers for dolphins – curved hooks, decoys, sinkers and night setting to protect albatrosses – biodegradable / radio-tracked traps to reduce ghost fishing • **Restricted effort:** limits on boat size, engine power and days at sea – decommissioning – restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • **Evaluation:** enforcement at sea is difficult – some methods reduce catch or raise costs – quotas need reliable stock data – effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate – as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **quota** → a legal limit on the mass or number that may be caught	9
	• **mesh size** → net hole size, set so that young fish escape • **biodegradable** → broken down by decomposers **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Fishing's impacts go beyond depleting target species: k-selected species that breed late and produce few young collapse and recover slowly; by-catch of immature target fish, over-quota fish and non-commercial species is thrown back dead, with drift nets killing whales, dolphins and turtles, longlines drowning albatrosses and pair trawls porpoises; lost gear keeps ghost fishing; demersal trawls destroy sea fans, corals and seagrass; and removing any species alters its competitors, prey and predators.	
	Catch quotas limit the weight landed and work well for single-species pelagic shoals where fishing stops when the quota is reached, but in mixed demersal fisheries over-quota fish are discarded dead, so the quota does not protect them. Gear design tackles by-catch directly: mesh sizes and square-mesh escape panels let small fish out because, unlike diagonal mesh, they do not close under drag; turtle exclusion devices, acoustic pingers for dolphins, curved hooks, decoys, sinkers and night setting protect turtles, cetaceans and albatrosses; biodegradable and radio-tracked traps limit ghost fishing. Restricting effort - boat size, engine power, days at sea, decommissioning - reduces total pressure, and banning drift nets or demersal trawling on sensitive seabeds such as Rockall Bank protects habitats.	
	Effectiveness depends on enforcement at sea, which is difficult, on reliable stock data for quotas, on the species and fishery type, and on international cooperation because fish migrate; some measures cut catches or raise costs, and as local stocks fall fleets simply move further away. Gear modifications are the most effective per unit cost because they reduce damage without reducing legal catch; quotas and effort limits only work where they are enforced.	
	Weaker answer - Level 2 (5 marks)	
	Fishing damages the environment by overfishing, catching by-catch such as dolphins and damaging the seabed with trawling. Quotas limit how many fish can be caught so stocks can recover. Bigger mesh sizes let small fish escape so they can grow and breed. Turtle exclusion devices let turtles escape from nets. Some fishing methods can be banned in sensitive areas. No-take zones let fish breed. These methods help but they are hard to enforce at sea and some fishermen ignore them.	
	What makes the difference	
	• Level 2 covers impacts and methods soundly but without the mechanisms (why square mesh works, why quotas fail in mixed fisheries).	
	• Level 3 evaluates each method on its own terms and then compares them; Level 2's evaluation is a single general sentence.	
	• Named examples (Rockall Bank, albatross measures, ghost-fishing controls) give Level 3 the range the top band expects.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Need:** population possibly 12 billion – humans raise carrying capacity – trophic level of diet • **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation – deforestation, fragmentation, management change – SSSIs, ancient woodland protection, coppicing, mixed planting • **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils – fuelwood, timber, agriculture, mining, reservoirs, climate change – protected areas, debt-for-nature swaps, sustainable exploitation • **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity – fisheries, erosion protection, medicines, tourism – damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species – MPAs, no-take zones, control of activities • **Deep-water reefs:** cold, dark, slow growth – research, fisheries – trawling, oil / gas, acidification – SACs • **Oceanic islands:** isolation, endemism, no mammal predators	25

Do not write outside the box

Question	Marking guidance	Marks
	– exploitation, introduced species, habitat change, sea level – eradications, control of development and visitors • **Mangroves:** halophytes, anaerobic mud – erosion protection, fisheries nurseries, sediment trapping – clearance for aquaculture / urban / ports, pollution, climate change – protection, replanting • **Antarctica:** extreme cold, seasonality, marine nutrients, ice – water store, albedo, carbon, research – climate change, ozone, tourism, fishing, minerals – Antarctic Treaty, CCAMLR, tourism and waste rules • **Synthesis:** features -> importance -> threats -> strategy – evaluation of success – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

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